PhishAlign Browser Extension — Setup Guide

What Was Built

A **Chrome/Edge Manifest V3** browser extension that:

- Intercepts every page navigation automatically
- Sends the URL to your **local FastAPI backend** (`localhost:8000/analyze`)
- The backend runs the **full pipeline**: Playwright page capture → URL model + HTML model + Visual model (MobileNetV3) + OCR + 25 consistency features → PhishAlign fusion
- **Hard-blocks phishing pages** with a full-screen overlay
- Shows the phishing probability + per-modality scores
- Lets the user override (with explicit acknowledgment) or go back to safety

Extension File Structure

```
extension/
├── manifest.json      — Chrome Manifest V3 config
├── background.js      — Service worker: intercepts nav, calls backend
├── content.js         — Content script: renders overlays on the page
├── popup.html         — Toolbar icon popup UI
├── popup.js           — Popup controller
└── icons/
    ├── icon16.png
    ├── icon48.png
    └── icon128.png
```

How to Load the Extension in Chrome / Edge

> IMPORTANT
>
> You need the **backend running** before loading the extension, otherwise scans will fail.

Step 1 — Start the FastAPI Backend

```
powershell
cd c:\Users\rohit\Desktop\projects\PhishAlign\backend
uvicorn app.main:app --host 127.0.0.1 --port 8000
```

Step 2 — Load the Extension

1. Open Chrome and go to: `chrome://extensions`
2. Enable **Developer mode** (toggle in the top-right corner)
3. Click **"Load unpacked"**
4. Select the folder: `c:\Users\rohit\Desktop\projects\PhishAlign\extension`
5. The PhishAlign shield icon will appear in your toolbar ✅

> NOTE
>
> For **Microsoft Edge**, go to `edge://extensions` and follow the same steps.

How It Works (Flow)

```
```

Popup States

State	Shown When
🔍 Scanning	Backend is processing the page
🚫 Phishing Detected	`prediction == "phishing"`
✅ Legitimate	`prediction == "legitimate"`
⚠️ Backend Offline	`localhost:8000` is unreachable

Hard-Block Screen Features

When phishing is detected, the content script shows:

- **Overall phishing probability** with animated progress bar
- **Per-modality breakdown**: URL / HTML / Visual probabilities
- **"Go Back to Safety"** button — navigates back in history
- **"Proceed anyway"** button — removes block (explicit user override)

Possible Issues & Fixes

Problem	Fix
All pages show "Backend Offline"	Start the FastAPI server on port 8000
Scanning takes 10–30 seconds	Normal — Playwright needs to launch a headless browser
Extension not seeing pages	Make sure Developer Mode is on and extension is reloaded after changes
`cors` errors in backend console	Backend CORS already updated to allow `chrome-extension://*`

CORS Update Made

— Added `allow_origin_regex` to allow requests from the extension:

```
python
allow_origin_regex=r"chrome-extension://.*",
```

Performance Note

> WARNING
>
> Each page visit triggers a **Playwright headless browser** launch in the backend (~5–15 seconds). This is by design (matches your research architecture using full page capture). For a production-

> grade extension, you'd add a URL allowlist/blocklist cache and only run the full pipeline on suspicious-looking URLs to reduce latency.